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Application of Geotechnical Instrumentation in Monitoring In-Pit Dykes Performance

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ABSTRACT

Open pit oil sands mining operations produce large quantities of tailings which are usually contained in external tailings facilities and in-pit dykes. At Muskeg River Mine, an extensive geotechnical instrumentation program is implemented as a part of design and construction control. This paper presents the design concept, planning and implementation of geotechnical instrumentation for in-pit dyke monitoring. The details of the pore pressure measurements, deformation monitoring and total stress monitoring are presented and discussed. The caution limits used for construction control are also presented and discussed.

RÉSUMÉ

L'exploitation de mines de sables bitumineux produit de grandes quantités de résidus qui sont contenus dans des dépôts externes et des digues dans des fosses. À la mine de la Rivière Muskeg, un vaste programme d'instrumentation géotechnique est mis en œuvre dans le cadre de la conception et de contrôle de la construction. Cet article présente la conception, la planification, et la mise en œuvre de l'instrumentation géotechnique pour surveiller la digue. Les détails des mesures de pression interstitielle et la surveillance de la déformation et de la contrainte totale sont présentés et discutés. Les limites d'avertissements utilisées pour le contrôle de la construction sont également présentées et discutées.

1 INTRODUCTION

Three major oil sands deposits (Cold Lake, Peace River and Athabasca) are located in the province of Alberta. Among these deposits, only the Athabasca deposit near Fort McMurray is shallow enough for surface mining operations. Surface mining produces large quantities of tailings which are stored within tailings containment structures. These structures are usually constructed with mine waste and coarse sand tailings. During the initial phase of the mining operation, tailings are contained in the above ground structures and as the mining progresses, tailings are contained within the mined out pit separated by In-Pit Dykes (IPDs).

Shell Canada Energy (SCE) operates Muskeg River Mine (MRM), which is one of the surface mining operations located about 70 km north of Fort McMurray, Alberta (Figure 1). Several in-pit tailings storage cells are currently being constructed at MRM, with five IPDs under construction. The locations of the tailings cells and the surrounding dykes are shown on Figure 2.

At MRM, the IPDs are designed utilizing the Observational Approach, where the design is based on mostly likely material parameters (Peck, 1969). Contingency plans are in place to allow safe construction in the event that less favourable parameters are operational in some areas of the fill or foundation. Successful application of observation approach requires a well-planned instrumentation program to confirm that the observed behavior is as expected or if not, to allow time to implement preplanned remedial measures.

The instrumentation program implemented at MRM includes installation of a variety of instruments, regular monitoring, data processing and interpretation. This paper presents the successful application of a geotechnical instrumentation program at MRM. The details of the pore

pressure measurements, deformation monitoring and total stress monitoring are presented and discussed. The caution limits used for design and construction control are also presented and discussed. The design and performance parameters reported in this paper represent the values used for the design of IPDs 1 and 2, at the time of writing. Design parameters are being updated as new data become available.

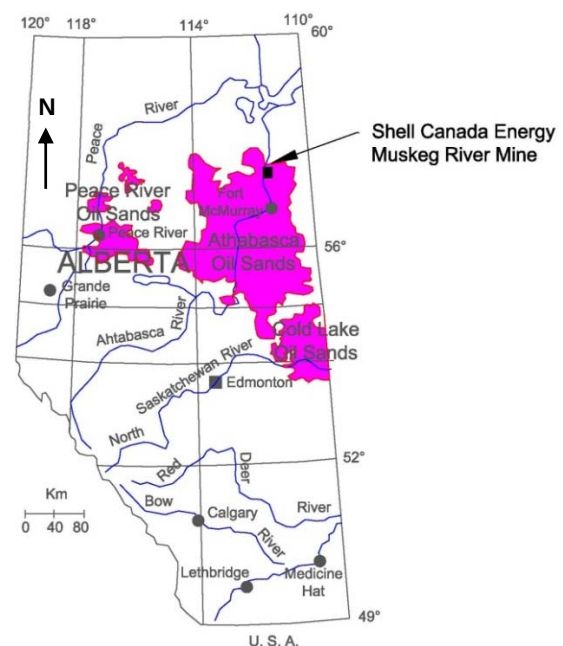


Figure 1 Location of Shell Canada Energy Muskeg River Mine Site

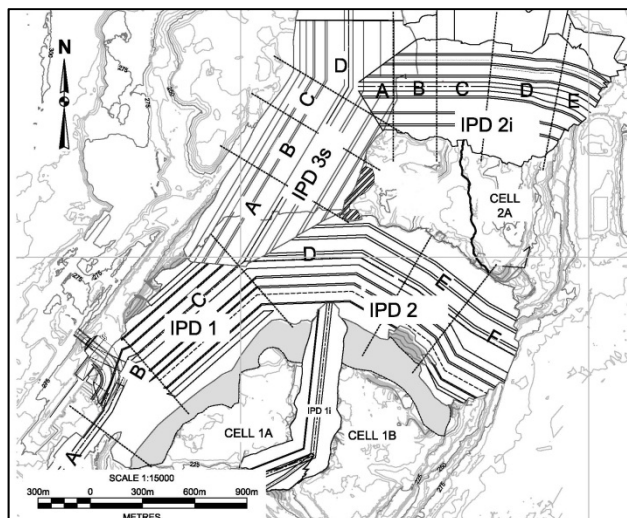


Figure 2 Plan view of MRM In-Pit Dykes

2 GEOLOGIC SETTING

The recent soil cover at MRM comprises Holocene and Pleistocene units including glacial till, glacio-fluvial, glacio-lacustrine and most recently fluvial and peat deposits.

The recent soils are underlain by the Cretaceous McMurray Formation containing layers of shale, sandstone, and bitumen rich sands. The McMurray Formation has been divided into three members; Upper McMurray (UM), Middle McMurray (MM) and Lower McMurray (LM). These members reflect rising sea level during deposition, grading upwards from a continental-fluvial, tidal/estuarine to near-shore marine environment.

The Middle and Lower McMurray are further divided into sub-members: MM1 and MM2, and LM1 and LM2, respectively. The ore body at MRM extends through all three McMurray Formation members. Although bitumen is present in each of the McMurray members, porous, clean sand deposits typically have the highest bitumen content. Water-bearing sands typically occur at the base of the Lower McMurray (LM1w) and are usually devoid of bitumen due to water saturation occupying the sand pore space. The waste units in each member are used as fill sources for dyke and other site construction.

The McMurray Formation is separated from the underlying calcareous shale and limestone of the Devonian Waterways Formation (DW) by a sub-Cretaceous unconformity. In some areas, the uppermost surface of the DW has weathered to a softened, high plasticity clay commonly referred to as Paleosol.

3 DESIGN CONSIDERATION

The Canadian Dam Association (CDA, 2007) classification for IPDs is considered "Very High". At MRM, IPDs are designed to satisfy the following key requirements:

- Provide the safe storage of tailings and fluids within the mined out pit;

- Satisfy regulatory requirements for Dam Safety (Alberta Environment) approvals and adhere to standard construction practices in the Oil Sands Industry;
- Meet the dyke construction schedule requirements for tailings containment;
- Simple and robust zonation that IPDs can be constructed with available mine waste materials utilizing mine equipment and personnel; and,
- Safely control of seepage flows.

To satisfy the varying foundation condition, IPDs are divided into sectors with similar foundation conditions (Figure 2). Depending on the foundation conditions and the Bitumen Production Plan, the dyke design may include shear keys, toe berms and seepage control measures such as chimney drains, downstream sand blankets together with granular French drains.

Quaternary overburden and most of the McMurray Formation materials were removed during the mining operation except for the abutments and waste islands where large bodies of ore-poor deposits are left in place. Otherwise, IPDs are usually founded on the LM1w unit or DW formation. The base of pit elevations generally ranged from 200 to 230 m above sea level (masl) with the original ground elevation of 290 masl, resulting in dykes that are 60 m to 90 m high.

4 FILL TYPE CLASSIFICATION

The fill materials used for dyke construction are: lean oil sands (containing less than 7% weight bitumen), clayey facies, and mixed facies that include bitumen rich layers that are too thin to isolate from the surrounding clays. Fill materials used in the initial construction of the MRM IPDs are divided into four groups as listed below:

- Type 1 - Coarser grained oil sand from MM1/MM2;
- Type 2 - Finer grained oil sand from MM1/MM2; and
- Type 5 - Overconsolidated clays typically from LM2; and,
- Trafficable Waste - any materials not meeting Types 1, 2 or 5 specifications but trafficable with a loaded Caterpillar 797 haul truck

Within each fill type, the engineering properties vary largely with the bitumen content, grain size, natural fluid content (defined as the ratio of mass of fluids (bitumen and water) divided by the mass of solids) and dry density to which the fill is compacted.

The dyke is divided into core, shell, and toe berm zones, with each zone having different material properties. Type 1 and 2 fill are allowed in all dyke zones. Type 1, 2 and Type 5 fills are allowed in the toe berms. Type classifications along with the minimum fluid content requirements and rut performance criteria are used as quality control measures to satisfy the dyke design requirements (Table 1).

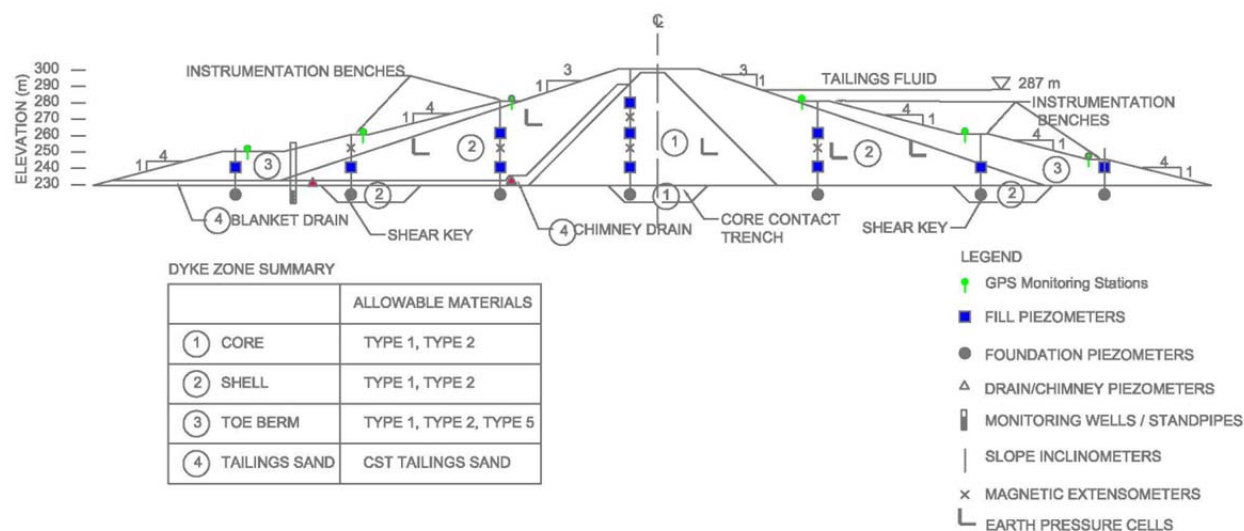


Figure 3 Typical cross section of IPD with instrument locations

Table 1 Fill type definition

Zone	Lift Thickness (m)	Min. Dry Density (kg/m ³)	Min. Fluid Cont. (%)	Max. Rut (mm)
Core (Type 1, 2)	1.0	1675	10 ⁺	200
Shell (Type 1, 2)	1.0	1700	8 ⁺	100
Toe Berm (Type 1, 2, 5)	1.5	1650	-	300

* Upon examination in a test trench, the fill should have no more 15% intact blocks and no visible voids.

A typical design section including different zones is presented in Figure 3. The maximum lift thickness and maximum rut specifications are based on compaction with loaded 797 haul trucks.

5 GEOTECHNICAL INSTRUMENTATION

5.1 Instrument Installation

The instrumentation program includes installation of Vibrating Wire Piezometers (VWPs), Slope Inclinometers (SIs), Magnetic Extensometers (MEs), Shape Accelerometer Array (SAA), Earth Pressure Cells (EPCs) and Standpipes (SPs). For each dyke, instruments are installed across a section of the dyke, usually at 200 to 300 m interval along the dyke's alignment. Generally, VWPs and SIs are installed using mud rotary drilling but augering is also used for shallower installations.

A typical instrumentation section is presented in Figure 3. VWPs are installed in the foundation, dyke fill, downstream sand blanket and chimney drain. SIs are installed in the both upstream and downstream. The base of SIs is seated in the stable foundation materials, usually in the DW Formation. MEs are installed together with SIs to monitor the settlement of the dyke fill. SAAs are

installed to monitor the deformation of the dyke fill and foundation material. Monitoring wells and SPs are installed in the basal water sands units to monitor the water level and water chemistry. In some locations, GPS monitoring stations are also installed to monitor the surface deformation of the dyke.

By taking advantage of installation during dyke construction, some of the fill VWPs and all of EPCs are installed in trenches using an excavator (Figure 4). Trench installation involves digging a 1 m deep x 1 m wide trench and installing the instruments at the base of the trench. After the installation, the trench is backfilled and instrument cables are trenched to a safe location for long term monitoring.



Figure 4 Trench Installation of Vibrating Wire Piezometer

Generally, most instruments are monitored on a bi-weekly basis, with more frequent readings of specific instruments based on the proximity to the construction activities and interpretation of the data. Geotechnical instruments utilized for IPD monitoring and their roles are summarized in Table 2.

Table 2 Summary of IPD instrumentation and roles

Instrument	During Dyke Construction	Long Term Monitoring
Fill Piezometers (VVPs)	Construction pore pressure – r_u response.	Phreatic surface. Potential for piping and hydraulic fracturing.
Foundation Piezometers (VVPs)	Construction pore pressure - $\bar{B}$ response.	Foundation pore pressure response to tailings pond. Potential for seepage/piping.
Drains/Chimney Piezometers (VVPs)	Monitor water level in the drains and chimney.	Monitor performance of the drains and chimney
Monitoring Wells/Standpipe (SP)	Monitor water level and water quality of basal water sands.	Same.
Slope Inclinerometer (SIs)	Monitor lateral deformation of the dyke fill and foundation.	Same.
Earth Pressure Cells (EPCs)	Monitor total stresses within the dyke fill.	Same.
Magnetic Extensometer (MEs)	Monitor vertical deformation of the dyke fill in specific zones.	Same.
Shape Accelerometer Array (SAA)	Monitor lateral and vertical deformation within the dyke fill and foundation.	Same.
Surface Markers/GPS monitoring Points	Monitor surface deformation.	Same.

5.2 Instrument Protection and Remote Monitoring Systems

One of the challenges at large mine sites like MRM is the protection and extension of the instruments. Since the dykes are being constructed as the mining progresses, the instruments need to be regularly extended up through the fill. When possible, instruments are trenched to the upstream and downstream instrumentation benches. Trenches are usually backfilled and compacted with a good quality fill material. To protect the instruments, instrumentation mounds are constructed around the instruments. Instrumentation mounds are usually 2 to 3 m high but can be up to 6 or 7 m high depending on the dyke construction schedule (Figure 5).

Vertical instrument extensions or horizontal trenches are not permitted in the core zone of any fluid-retaining dyke, to prevent the formation of defects in the core and the potential for concentrated seepage. Instruments in the core zone may only be installed in holes drilled from the constructed dyke surface.

Proper placement and compaction around the mounds is very important to ensure a loosely compacted vertical zone does not occur in the dyke. Where instruments or instrument leads are exposed at the surface of a mound or the dyke surface, painted wooden frames (referred to

as quads) are used to mark and protect them from damage.



Figure 5 Instrument Protection Mound

With the advancement of remote monitoring systems, more instruments are remotely monitored. At MRM, some of the instruments are monitored using data loggers and remote monitoring systems (Cell and Radio modems). The advantage of remote monitoring systems is that the data can be downloaded from the office without requiring field staff to take the readings.

6 PORE PRESSURE MEASUREMENT

6.1 Construction Pore Pressure (Fill)

Pore pressure generated during the embankment dyke construction has been widely studied (Bishop, 1957; Penman, 1979; Skempton, 1954). It is acknowledged that construction pore pressure is a function of placement water content, drainage conditions, fill height, fill permeability, degree of compaction and rate of construction.

In traditional embankment dam construction, fill water content is controlled to be near Standard Proctor Maximum Dry Density (SPMDD) to achieve higher compaction, and the fill is compacted to a relatively high degree of saturation. Due to that, the traditional embankment dams can develop high construction pore pressure during the dyke construction.

In the oil sand industry, IPDs are constructed using fill material with the natural total fluid content (moisture + bitumen) which is usually less than SPMDD optimum moisture content, as shown on Figure 6. Compaction is achieved using fully loaded Caterpillar 797 haul trucks (total weight ~ 700 ton). Consequently, most of the dyke fills tend to have a higher dry density compared to SPMDD results.

Figure 7 shows typical Troxler field density versus fluid content measurements taken during the core, shell and toe berm construction together with the zero air voids line and saturation lines. As shown in the figure, most of the measured points are located between the 60% and 80% saturation lines.

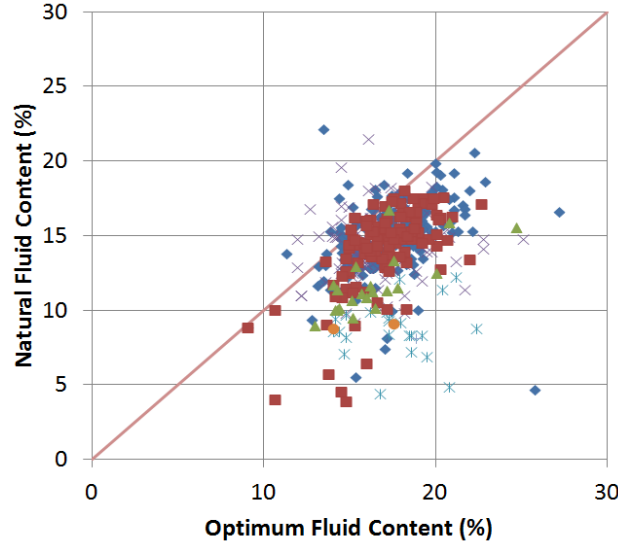


Figure 6 Comparison between field and optimum fluid contents for compacted lean oilsand fills

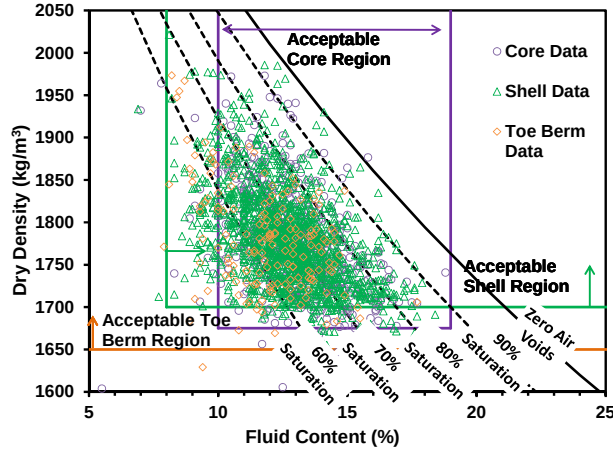


Figure 7 Troxler field dry densities measured during dyke construction

Pore pressure developed in the dyke fill during construction is usually presented as the pore pressure ratio, r_u which relates the pore water pressure to fill above the piezometer tip. The pore pressure ratio is defined as:

$$r_u = \frac{u}{\gamma_t H_s} \quad [1]$$

where:

u = the pore water pressure

γ_t = the total unit weight of the soil

H_s = the height of the soil column

Pore pressure ratios of fill materials used for the initial IPD design are presented in Table 3. At MRM, most of the fill piezometers registered r_u values in the range of 0.0 to 0.25 which is normal for fill compacted dry of Standard Proctor optimum moisture content. During the dyke construction, some fill piezometers show increases in pore pressure responding to fill placement. However once

the dyke construction moves to other locations, piezometric pressures decrease with time. The silty nature of the dyke fill and the low permeability of the sandier materials due to the presence of minor amounts of bitumen result in moderate rates of dissipation of the construction pore pressures. This type of behaviour is presented in Figure 8.

Table 3 Fill materials pore pressure ratio design values

Material Zone	Fill Type	Design r_u Value
Core	Type 1, Type 2	0.5
Shell	Type 1, Type 2	0.3
Toe Berm	Type 1, Type 2, Type 5	0.4

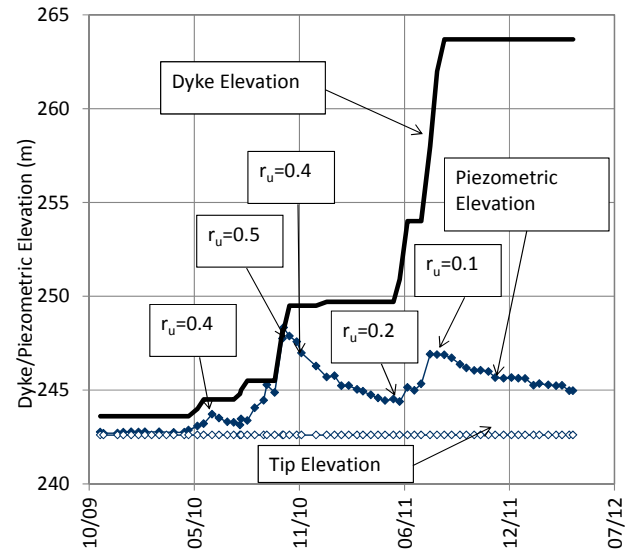


Figure 8 Pore pressure response of fill piezometer due to dyke construction

As shown in Figure 8, construction pore pressure increases as the dyke construction progresses in the area. Initially, pore pressure increase in response to the fill placement is high, registering a r_u value of 0.5. When the dyke construction stops, the excess pore pressure generated in the fill starts to dissipate with time. With each increment of dyke elevation, pore pressure increase becomes less pronounced and the r_u value reduced below the design limit. This observation is in agreement with the pore pressure monitoring data observed in other embankment dams (Bishop, 1957).

Some fill piezometers are registering negative pore pressures, ranging from 0.0 kPa to 35 kPa. Penman (2002) indicated that negative pore pressure can be developed in partly saturated fill due to compaction, a scenario that has been observed in some of the VWP's in the fill material.

Some fill piezometers registered high pore pressures with r_u ratios above the design value. These high r_u piezometers are generally associated higher fluid content fill (Seto et al., 2009). Figure 9 presents the measured r_u values of the fill piezometers with respective fill height

above the piezometer tip. As shown, most of the fill piezometers fall below the r_u value of 0.3. This figure aggregates data with different 'ages'; some locations have had more dissipation time than others.

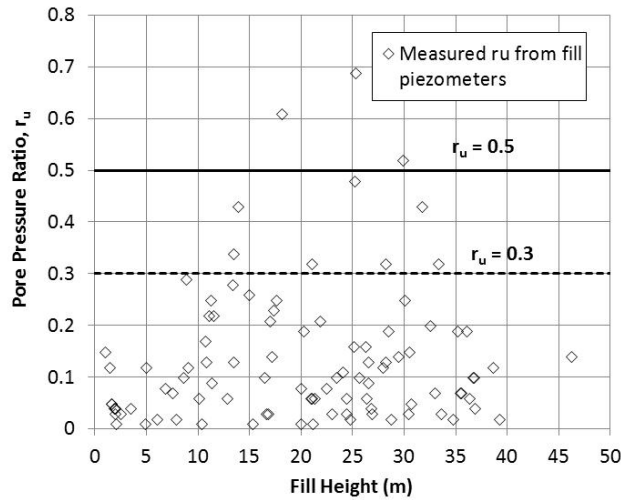


Figure 9 Pore pressure ratio, r_u vs. fill height (m)

Comparing the data in Figure 9 to the design parameters listed in Table 3, it is evident that most of the measured values are below the design r_u values. The initial design parameters were based on experience with fills under lower total stresses that exhibited relatively high pore pressure response. The data obtained from monitoring the construction of these dykes allowed r_u values to be selected that are appropriate to the stress range for full height design, and resulted in an overall improvement in the fill utilization of the designs.

6.2 Foundation Pore Pressure

Understanding pore pressure response of the foundation material is also a major component of the dyke design and construction monitoring. At MRM, the groundwater level is located just below the original ground surface prior to mining in the area. As the mining proceeds, the pit bottom is progressively dewatered. Due to the excavation and dewatering, piezometers installed behind the pit wall show decreases in piezometric elevations during the mining phase.

Once the mining is completed and the area is cleared, construction of IPD can proceed. The pore pressure response of the foundation due to dyke construction can be represented as pore pressure ratio $\bar{B}$, which is defined as the ratio of the change in pore pressure to the incremental change in fill pressure as described below:

$$\bar{B} = \frac{\Delta u}{\Delta \sigma_v} \quad [2]$$

where:

Δu = change in pore pressure

$\Delta \sigma_v$ = change in vertical stress

Pore pressure ratios of foundation materials used for IPD design are presented in Table 4. To monitor the pore pressure response due to dyke construction and tailings deposition, several VWP's were installed in the MM and LM Members, basal water sand units (LM1w) and DW Formation. Due to the high permeability of the LM1w, piezometers installed in the LM1w do not show any response to dyke construction. But LM1w piezometers do show response to the fluid levels in the tailing cells.

Table 4 Foundation materials pore pressure ratio design values

Material Zone	Design $\bar{B}$ value
MM1/LM2	0.4
Devonian calcareous shale and limestone	0.3
Devonian paleosol	0.4

Most of the piezometers installed in the MM, LM and DW formations show minimal response to dyke construction. This low $\bar{B}$ response can be attributed to stress relief caused by excavation of up to 90 m of material prior to dyke construction. The pore pressure response in the foundation materials is highly complex due to unloading and reloading. It is possible that a higher $\bar{B}$ response will be observed as the dyke elevation reaches above the original ground level. The pore pressure response of the DW paleosol is generally higher than for the underlying, unweathered calcareous shale and limestone. Measured $\bar{B}$ values in the Paleosol typically ranged from 0.1 to 0.3. $\bar{B}$ values of MM and LM piezometers generally ranged from 0.05 to 0.15. Figure 10 presents the measured $\bar{B}$ values of the foundation piezometers with respective fill height above the piezometer tip.

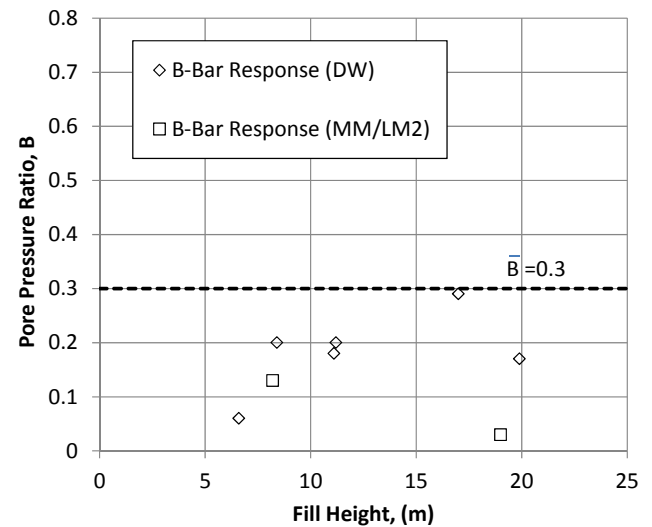


Figure 10 Pore pressure ratio, $\bar{B}$ vs. fill height (m)

7 DEFORMATION MONITORING

7.1 Horizontal Movement

Monitoring frequency of SIs generally ranged from bi-weekly to monthly but during nearby dyke construction the monitoring frequency is increased depending on the observed deformation. If a higher rate of movements is observed, monitoring frequency is increased to weekly and sometimes daily.

For the IPDs, the observed movements are associated with the dyke construction. As the dyke construction moves away from the area, the rate of movement starts to slow down and eventually stop. Most of the observed lateral movements ranged from 0.01 to 0.07 mm/day. However, during the active nearby dyke construction, movement rates have gone as high as 0.2 to 3.0 mm/day. The majority of the movement is associated with bulk deformation and settlement, with distributed strains over 30 m to 40 m vertical zones in the dyke fill. Discrete shear zones have not been observed in the IPD fill or foundations at MRM.

A typical pattern of lateral deformation of a SI is presented in Figure 11. The left figure shows the cumulative movement of the inclinometer at two different times and the right figure shows the time vs. displacement plot of the zone of movement (in mm at primary Y axis) together with the fill height (in m at secondary Y axis). As shown, very little movement is observed when there is no nearby dyke construction. The rate of movement starts to increase once the dyke construction starts in the area, and the rate of movement reduces quickly when the construction moves away from the area.

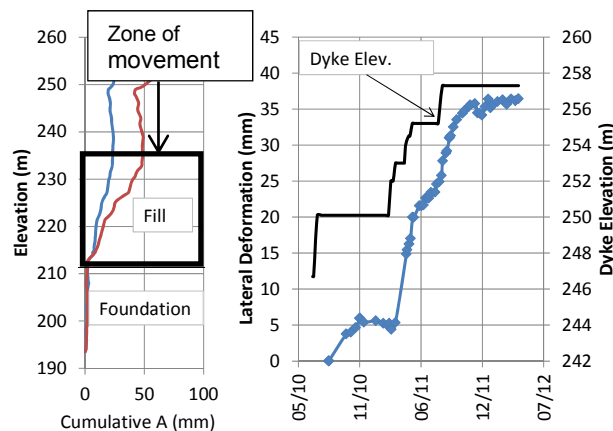


Figure 11 Lateral deformation due to dyke construction

To determine appropriate responses to various dyke displacements, a series of caution limits have been established as shown in Table 5. The caution limits are divided into four levels (from Green to Red) depending on the observed rate of movements and total shear strains; these criteria are independent of one another. If the observed movement rate is within the green range, normal construction and monitoring proceed. If the rate of movement reaches the yellow and orange ranges, monitoring frequency is increased. If the rate of

movements reaches to red range, dyke construction in the area of movement is temporarily stopped until the rate of movement is reduced to an acceptable level.

Table 5 Caution limits established for in-pit inclinometers

Rate of Movement(m m/day)	Total Shear Strain(%)	Caution Levels	Monitoring Schedule
< 0.07	< 0.5	Green	Bi-Weekly
> 0.07 to < 0.2	> 0.5 to < 1.0	Yellow	Weekly to Twice/Week
> 0.2 to < 1.0	> 1.0 to < 2.0	Orange	Twice a week to Daily
> 1.0	> 2.0	Red	Daily

7.2 Vertical Movement

Seismic surveys in the vicinity of IPD 2i indicated the possible presence of a north-south trending fault in the DW Formation, crossing the axis of IPD 2i near the mid-point of the dyke. Faulting in the DW Formation is understood to result from dissolution of halite beds in the Prairie Evaporite Formation and collapse of the overlying strata, roughly coincident with deposition of the LM Member. Drilling in the area has shown that the dissolution of the halite beds is complete and the overlying material has fully collapsed; hence, no further large-scale movement along this potential fault would be expected. Nonetheless, a horizontal Shape Accelerometer Array (SAA) cable was placed in a trench parallel to the axis of IPD 2i, immediately downstream of the chimney drain, to check if minor movements would occur along this potential fault due to the weight of the fill placed in Dyke 2i. No movement has been detected along this potential fault during the placement of the first 40-50 m of fill in IPD 2i.

8 TOTAL STRESS MONITORING

Total stress monitoring was incorporated into the first IPDs to obtain a better understanding of the 3D stress conditions within the compacted fills. During the dyke construction, low stress zones can be developed in the fill where the foundation elevation changes rapidly or near the steep abutments. These low stress zones are susceptible to hydraulic fracturing.

To measure the total stresses developed in the dyke fill, Earth Pressure Cells (EPCs) were installed near the base of the fill and near the abutment tie-in with the dyke fill. To monitor both vertical and horizontal stress, two EPCs (one horizontal and one vertical) were installed in each location together with a VWP to monitor the pore pressure.

The EPCs installed in the IPDs have shown good total stress response. EPCs register increases in pressure during the dyke construction and the pressure stabilizes when the construction moves away from the area (Figure 12). Small decreases in pressure were observed after the dyke construction moved away from the area which can

be attributed to dissipation of the construction pore pressure in the dyke fill.

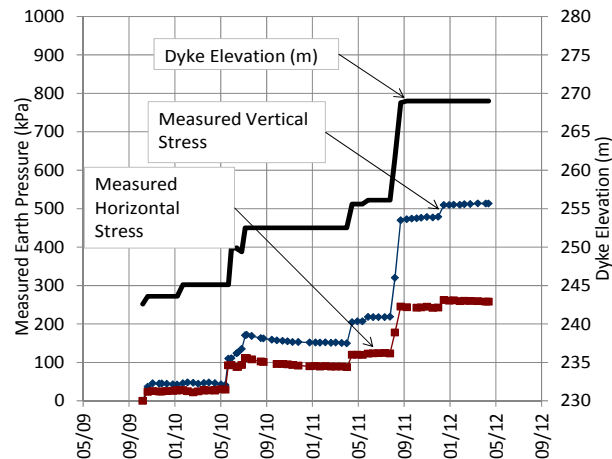


Figure 12 Graphical plot of measured total vertical and horizontal stress together with constructed dyke elevation

Figure 13 shows the measured vertical and horizontal total stresses at different depths. The measured total vertical stress, σ_v , is very close to the calculated overburden stress (assuming a unit weight of 19.5 kN/m^3), and the measured total horizontal stress, σ_h is close to 50% of σ_v . Using the horizontal and vertical total stresses measured with the EPCs in conjunction with the pore pressures, u , measured in adjacent VWP's, the coefficient of lateral earth pressure at rest (K_o) can be determined using the following relationship:

$$K_o = \frac{\sigma_h - u}{\sigma_v - u} \quad [3]$$

For the IPD fill, measured K_o values deduced from EPCs ranged from 0.4 to 0.7. The K_o data were used to assess the potential for hydraulic fracture of the dykes and confirm the suitability of the defensive measures in the dyke design, using the methods outlined by Cameron et al. (2008). A plot of K_o vs. Depth is presented in Figure 14.

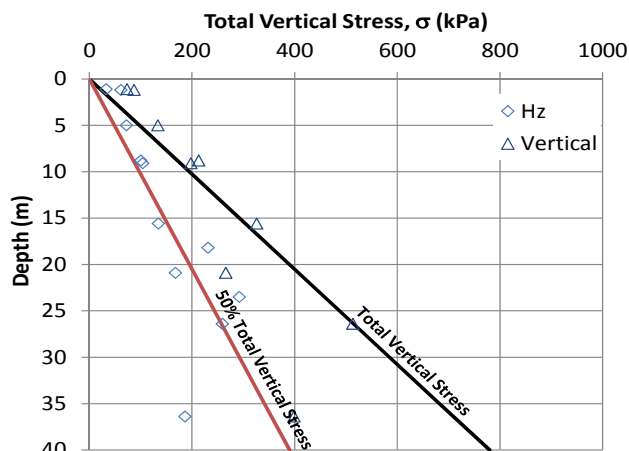


Figure 13 Measured horizontal stress from EPCs and pressuremeter tests

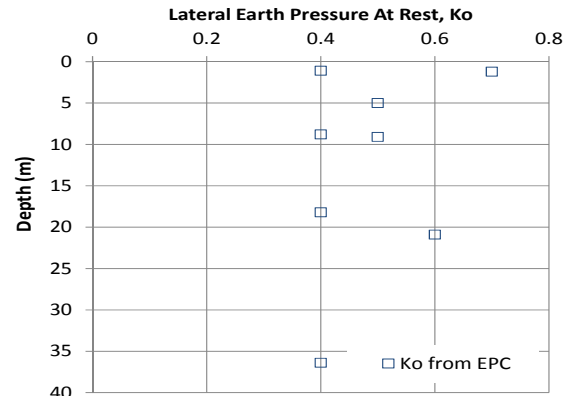


Figure 14 Lateral Earth Pressure at Rest, K_o vs. Depth

9 CONCLUSIONS

An extensive geotechnical instrumentation program is implemented at the MRM site to monitor the IPD performance, in support of the Observational Approach to dyke construction. This approach has allowed for safe construction of the dykes and optimization of several key design parameters.

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